

CHIANG MAI INTL, Thailand

WMO: 483270

Lat: 18.7714N

Lon: 98.9692E

Elev: 314

StdP: 97.61

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.1	(c) 14.5	(d) 6.0	(e) 6.0	(f) 29.6	(g) 8.0	(h) 6.9	(i) 28.0	(j) 5.4	(k) 24.8	(l) 4.5	(m) 26.0	(n) 0.3	(o) 260	(p) 0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.6	38.1	22.8	36.9	22.7	35.8	22.6	26.9	31.9	26.4	31.4	26.0	30.9	2.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	22.0	29.5	25.2	21.1	28.3	24.9	20.8	28.0	87.1	32.2	84.8	31.5	83.0	31.3	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.1	4.3	10.6	39.4	2.3	1.3	8.9	40.4	7.6	41.2	6.2	41.9	4.6	42.9		
			DB	9.5	28.0	2.1	1.0	8.0	28.7	6.7	29.2	5.6	29.8	4.0	30.5	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	22.8	24.9	27.7	29.9	29.0	28.5	27.9	27.5	27.6	27.0	25.3	23.0
	DBStd	2.76	1.79	1.83	1.85	1.89	2.04	1.34	1.37	1.20	1.16	1.33	1.91	2.24
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	1	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6119	397	417	547	598	589	556	554	543	528	528	458	403
	CDD18.3	3081	140	183	289	348	331	306	296	285	278	270	208	147
	CDH23.3	31237	1154	1924	3459	4498	3776	3275	2939	2618	2506	2366	1664	1058
CDH26.7	13675	400	941	1923	2580	1853	1339	1057	862	884	897	616	322	
Wind	WSAvg	1.4	0.9	1.1	1.4	1.8	1.9	1.8	1.6	1.5	1.4	1.2	1.1	1.0
Precipitation	PrecAvg	1171	9	8	18	53	165	131	156	224	218	125	51	12
	PrecMax	1577	79	66	181	214	411	336	446	425	530	249	332	77
	PrecMin	738	0	0	0	1	13	15	52	67	64	13	0	0
	PrecStd	218	18	15	31	46	87	63	74	85	88	62	62	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.3	35.8	38.1	40.0	39.1	36.2	35.1	33.9	34.2	34.1	33.2	32.1
		MCWB	20.3	20.1	21.2	22.4	24.0	25.3	25.3	25.5	25.6	25.1	23.5	22.5
	2%	DB	31.2	34.3	37.0	38.8	37.0	34.8	33.7	32.8	33.1	33.0	32.2	30.9
		MCWB	20.0	19.7	20.9	22.7	24.5	25.3	25.1	25.3	25.2	24.6	23.2	21.7
	5%	DB	30.2	33.2	36.0	37.6	35.3	33.2	32.4	31.9	32.1	32.1	31.2	29.9
		MCWB	19.7	19.4	20.6	22.7	24.7	25.2	25.0	25.1	25.1	24.3	22.9	21.2
	10%	DB	29.1	32.1	34.9	36.1	34.0	32.1	31.2	30.9	31.1	31.1	30.2	28.8
		MCWB	19.4	19.4	20.5	23.0	24.7	25.1	24.8	25.0	24.9	24.2	22.5	20.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.0	23.4	24.6	26.6	27.4	28.1	27.2	27.1	27.2	26.5	25.7	24.4
		MCDB	28.6	31.4	33.0	34.7	34.0	32.3	31.9	31.0	31.5	31.4	30.5	29.5
	2%	WB	22.1	22.3	23.6	25.6	26.5	26.7	26.2	26.4	26.4	25.8	24.7	23.3
		MCDB	27.4	29.9	32.2	33.9	33.0	31.6	30.7	30.4	30.6	30.7	29.6	28.3
	5%	WB	21.4	21.6	23.0	24.9	25.9	26.1	25.7	25.9	25.9	25.3	24.1	22.5
		MCDB	27.0	29.1	31.7	33.2	32.1	31.0	30.0	29.8	30.0	30.0	28.9	27.2
	10%	WB	20.7	20.8	22.3	24.2	25.4	25.6	25.4	25.5	25.5	24.8	23.6	21.7
		MCDB	26.6	28.6	30.4	32.1	31.2	30.1	29.3	29.1	29.3	29.2	28.0	26.2
Mean Daily Temperature Range	5% DB	MDBR	13.8	15.5	14.7	12.6	9.6	8.1	7.2	7.1	7.8	8.9	10.4	11.8
		MCDBR	15.1	16.8	16.2	14.3	11.7	9.6	8.9	8.5	9.1	10.1	11.2	12.5
		MCWBR	5.7	5.7	5.5	4.7	4.2	3.5	3.2	3.3	3.4	3.6	4.1	4.8
	5% WB	MCDBR	12.4	14.0	13.4	12.3	10.1	8.6	7.7	7.6	8.2	9.0	9.8	10.7
		MCWBR	5.1	5.3	5.0	4.6	3.9	3.5	3.1	3.2	3.4	3.5	3.8	4.3
Clear-Sky Solar Irradiance	taub	0.434	0.510	0.679	0.677	0.507	0.442	0.429	0.445	0.476	0.515	0.441	0.436	
	taud	2.206	2.001	1.619	1.687	2.162	2.380	2.413	2.351	2.243	2.092	2.262	2.227	
	Ebn at Noon	840	793	678	685	803	850	862	854	828	779	828	825	
	Edh at Noon	136	175	265	250	154	122	119	127	140	158	127	129	
All-Sky Solar Radiation	RadAvg	4.62	5.41	5.66	6.00	5.35	4.64	4.30	4.26	4.52	4.44	4.58	4.32	
	RadStd	0.23	0.19	0.36	0.35	0.46	0.32	0.33	0.26	0.19	0.24	0.33	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.66	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	+271	+273
Regional (15 neighbors)	+0.28	N/A	N/A	N/A	N/A	+0.15	N/A	N/A	+138	+139

Nomenclature: See separate page